Date: 22/06/2026

Lab Practical #02:
Study of different network devices in detail.

Practical Assignment #02:
1. Give difference between below network devices.
   - Hub and Switch
   - Switch and Router
   - Router and Gateway
2. Working of below network devices:
   - Switch
   - Router
   - Gateway

Hub and Switch

No.	Hub	Switch
1	A hub is a basic networking device that connects multiple computers.	A switch is an intelligent networking device that connects multiple devices efficiently
2	Works on the **Physical Layer (Layer 1)** of the OSI model.	Works on the **Data Link Layer (Layer 2)** of the OSI model
3	Sends data to **all connected devices** (broadcast).	Sends data only to **the intended device** using its MAC address.
4	Slower because all devices share the same bandwidth.	Faster because each device gets dedicated bandwidth.
5	More network traffic and collisions occur.	Very few collisions because it creates separate collision domains.

Switch and Router

No.	Switch	Router
1	A switch connects multiple devices within the same Local Area Network (LAN).	A router connects different networks (e.g., LAN to the Internet).
2	Works on the **Data Link Layer (Layer 2)** of the OSI model.	Works on the **Network Layer (Layer 3)** of the OSI model.
3	Uses **MAC addresses** to forward data.	Uses **IP addresses** to route data.
4	Creates separate **collision domains** for each port.	Creates separate **broadcast domains**.
5	Faster for local network communication.	Slightly slower than a switch because it performs routing decisions.

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Router and Gateway

No.	Router	Gateway
1	A router connects **different networks**.	A gateway connects **different networks with different protocols**.
2	Works on the **Network Layer (Layer 3)** of the OSI model.	Can operate at **all layers (Layer 1–7)** depending on its function.
3	Uses **IP addresses** to forward data packets.	Performs **protocol conversion** so different networks can communicate.
4	Routes data by finding the best path between networks.	Translates data between different communication protocols or formats.
5	Mainly used to connect a LAN to another LAN or the Internet.	Used to connect networks using different technologies or protocols.

Working of below network devices:

1. Switch
    - A device sends a data packet to the switch.
    - The switch checks the **MAC address** of the destination device.
    - It looks for this MAC address in its **MAC address table**.
    - If the address is found, the switch sends the data **only to the correct device**.
    - If the address is not found, the switch sends the data to all devices once. After receiving a reply, it learns the MAC address and stores it in its table for future communication.

2. Router
    - A device sends a data packet to the router.
    - The router checks the **destination IP address** of the packet.
    - It looks at its **routing table** to find the best path to the destination.
    - The router forwards the packet through the selected route.
    - If the destination is on another network (such as the Internet), the router sends the packet to the next router until it reaches the destination.

3. Gateway
    - A device sends data to the gateway.
    - The gateway receives the data packet.
    - It checks whether the source and destination networks use different protocols or formats.
    - If needed, the gateway **converts (translates)** the data into a format that the destination network can understand.
    - The gateway then forwards the converted data to the destination network.